# MamNet-6: Multi-View Fusion, CrossView Transformer, and Uncertainty-Aware Triage for Breast Cancer Detection on VinDr-Mammo

Haoming Luo, Matthew Hamilton, Oscar Meruvia-Pastor, and Edward Kendall
*Department of Computer Science*
*Memorial University*
St. John's, NL, Canada

***Abstract*—Most published work on the VinDr-Mammo benchmark processes one view at a time, ignoring the CC/MLO pairing that radiologists rely on. We address this by building MamNet-6, a six-stage pipeline that progressively adds multi-view fusion, spatial cross-attention, uncertainty quantification, and hybrid triage simulation, and we validate it zero-shot on the RSNA 2022 dataset. DenseNet-121 achieved an AUROC of 0.756 as a single-view baseline. BreastPairNet, a shared-backbone network that jointly processes CC and MLO views, brought this to 0.826 [95% CI 0.776–0.871] and the CrossView Transformer to 0.832 [0.783–0.875]; individual stage gains sit within overlapping confidence intervals, so the cumulative improvement is the more reliable comparison. The 16-model ensemble with test-time augmentation achieved an AUROC of 0.846 (an 11.8% gain over single-view CNN) with sensitivity at 90% specificity of 0.657 and PPV of 0.184 at the triage threshold. Hybrid triage at 50% AI reads preserved 87.9% sensitivity under oracle conditions. Two unexpected findings stand out: bounding-box annotations degraded AUROC by 0.12–0.20, suggesting the model learns to exploit box-region features rather than the underlying pathology, and zero-shot transfer to RSNA 2022 yielded an AUROC of 0.601, pointing to a real domain gap between Vietnamese and North American screening populations.**

***Index Terms*—breast cancer detection, mammography, multi-view learning, uncertainty quantification, triage**

## I. Introduction

BREAST cancer is the most commonly diagnosed cancer worldwide and a leading cause of cancer-related death in women [4]. Population screening programmes based on full-field digital mammography (FFDM) have repeatedly been shown to reduce breast-cancer mortality, yet they impose a heavy operational burden: in many countries each screening exam is double-read by two radiologists, and reader sensitivity varies by 10–30 percentage points between individuals [10], [11]. False-negative rates of 10–30% and false-positive recall rates of approximately 8% are well documented, and historical hand-engineered detection systems failed to deliver measurable benefit in routine screening [17].

Deep learning has changed this: CNNs trained on hundreds of thousands of exams now reach or exceed individual radiologist AUC on internal and some external cohorts [11], and meta-analyses report pooled AUC of 0.87 for AI on digital mammography versus 0.81 for radiologists [4]. Three trends are particularly relevant. First, multi-view modelling that fuses the craniocaudal (CC) and mediolateral oblique (MLO) projections improves over single-view inference because findings often appear unambiguously in only one view [10]. Second, transformer-based fusion of views allows feature-level interaction rather than score-level averaging. Third, uncertainty-aware inference via Monte Carlo (MC) Dropout [13] enables a *hybrid triage* protocol in which the AI reads the most-confident exams autonomously and refers the rest to a human reader [12], reducing radiologist workload by approximately 38% in retrospective evaluation [12].

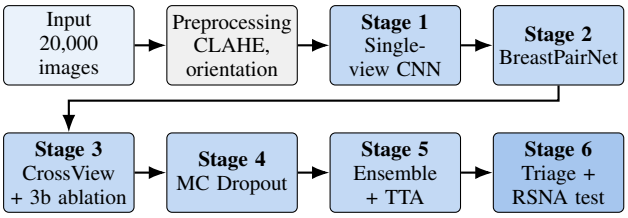



Fig. 1. MamNet-6: overview of the six-stage breast cancer detection pipeline.

The VinDr-Mammo benchmark [1] is a 20,000-image, patient-level, 4-view FFDM dataset. Three gaps in the literature are evident. **(i) Single-view dominance.** Most published VinDr-Mammo results train one image at a time. **(ii) Spatial cross-view attention is underexplored.** Existing multi-view designs typically concatenate global features. **(iii) Uncertainty quantification and triage simulation are rare.** No published study has applied MC Dropout to VinDr-Mammo and simulated a Verboom-style triage policy [12].

We address these gaps with MamNet-6, a six-stage pipeline (Figure 1).

## II. Related Work

**CNN backbones.** Residual [7] and densely connected [8] networks underpin most screening AI systems; CheXNet [6] made DenseNet-121 a default starting point for chest imaging, EfficientNet [9] introduced compound scaling, and ConvNeXt [16] brought pure-convolutional design to transformer-level ImageNet accuracy.

**Multi-view mammography.** Wu et al. [10] trained a four-stream ResNet-22 on 229,426 NYU exams (AUC=0.895), showing that shared backbones and feature-level aggregation

matter under limited data; McKinney et al. [11] reported AUC=0.889 on UK FFDM, outperforming six radiologists.

**VinDr-Mammo.** Nguyen et al. [1] released VinDr-Mammo with BI-RADS, density, and lesion-level annotations. MamT4 [18] reports AUC=0.840±0.017; Abdikenov et al. [2] showed CLAHE+YOLOv12 mAP gains, and Mellado et al. [3] competitive patch-level F1 under limited annotation.

**Imbalance and uncertainty.** Focal loss [5] down-weights easy negatives by $(1-p)^{\gamma}$, and Johnson and Khoshgoftaar [15] survey algorithm-level imbalance methods. Kendall and Gal [13] distinguish aleatoric from epistemic uncertainty; Verboom et al. [12] reduced workload by 38.1% on 41,469 Dutch screening exams while preserving cancer detection rate.

## III. Datasets

### A. VinDr-Mammo (primary dataset)

VinDr-Mammo [1] is a Vietnamese FFDM benchmark containing 20,000 images from 5,000 patients, four views per study (R-CC, L-CC, R-MLO, L-MLO) stored as DICOM, with BI-RADS (1–5), density (A–D), and lesion-level bounding box annotations (Table I).

TABLE I
BI-RADS distribution in VinDr-Mammo and binary label mapping.

| BI-RADS | Images | Binary label |
|---|---|---|
| 1 | 14,036 | negative |
| 2 | 4,144 | negative |
| 3 | 832 | negative |
| 4 | 824 | **positive** |
| 5 | 164 | **positive** |
| **Total** | **20,000** | 988 positive (4.94%) |

The official split is 4,000 training patients (16,000 images) and 1,000 test patients (4,000 images). For multi-view experiments we construct one ipsilateral CC+MLO pair per breast, yielding 8,000 training pairs and 2,000 test pairs, of which 99 test pairs (4.95%) are cancer-positive. A 5% validation slice (˜400 pairs) is held out for early stopping; the official test split is never used for training or checkpoint selection.

### B. RSNA 2022 (external dataset)

The RSNA 2022 dataset comprises 54,706 screening DICOMs from 11,913 patients, normalised as for VinDr-Mammo (about 29,519 JPEG Lossless files decoded with `pylibjpeg`). Excluding non-CC/MLO views leaves 23,826 CC+MLO pairs (19,062 training, 4,764 validation; 80/20 patient-level split).

## IV. Methods

### A. DICOM Preprocessing

Four steps: (1) decode with pydicom+`pylibjpeg`; (2) percentile-clip to $[1\%, 99\%]$; (3) CLAHE [14] (32×32 tiles, clip-limit 2.0); (4) standardise breast orientation so that the breast faces right—a horizontal flip is applied when the mean intensity of the left image half exceeds the right, as breast tissue is brighter than the background. Output: single-channel 8-bit PNG replicated to 3 channels.

### B. Single-view CNN Backbones (Stage 1)

Five configurations: **VGG19**, **ResNet-50** (224 px), **ResNet-50** (384 px), **EfficientNet-B3** (224 px), **DenseNet-121** (224 px). All use ImageNet weights, focal loss ($\alpha$=0.25, $\gamma$=2.0) [5], WeightedRandomSampler, and AdamW with cosine decay. Training: 20 epochs with early stopping on validation AUROC.

### C. BreastPairNet (Stage 2)

Shared backbone produces per-view global-average-pooled vectors $\mathbf{f}_{CC}, \mathbf{f}_{MLO} \in \mathbb{R}^d$. Two fusion heads: **concatenation** (2d-dim MLP, dropout 0.5) and **attention** (learned per-view scalar weights via softmax). The **+reg** variant adds stronger elastic deformation (`alpha`=50, `sigma`=5), mixup ($\alpha$=0.2), and label smoothing (0.1). Training: backbone frozen 3 epochs, then uniform learning rate 1e-4 applied to all parameters. Figure 2 illustrates the architecture.

### D. CrossView Transformer (Stage 3)

Replaces global-pooled fusion with spatial cross-attention. Both views pass through the shared CNN backbone to the last spatial feature map; spatial tensors are flattened to tokens and entered into a two-layer Transformer encoder where queries from each view attend to keys/values of the other. Cross-attended outputs are global-averaged and concatenated for the final classifier. Figure 3 contrasts this with BreastPairNet.

### E. Bounding-box Annotation (Stage 3b)

Ground-truth boxes (VinDr-Mammo) were drawn as white outlines (40-px line width) on training images only. For RSNA 2022, a YOLOv8s detector pre-trained on VinDr-Mammo was used (0.8% sensitivity; noisy hints only). Test sets always used original unannotated PNGs to prevent label leakage.

### F. MC Dropout (Stage 4)

MC Dropout [13] keeps dropout active at test time, turning a single prediction into a distribution without architectural changes or retraining, so entropy-based case ranking costs little. BatchNorm is kept in eval mode while dropout remains active. $N = 20$ stochastic forward passes per test pair yield probabilities $\{p_i\}_{i=1}^{20}$; their mean $\bar{p}$ is used for classification and the binary entropy $H(\bar{p}) = -\bar{p}\log\bar{p} - (1-\bar{p})\log(1-\bar{p})$ quantifies predictive uncertainty. The classification threshold is the Youden-optimal cut-off, maximising Youden's J index ($J$ = sensitivity + specificity − 1) on the evaluation set; the same rule is used for every threshold reported in this paper.

### G. Hybrid Triage (Stage 6)

Following Verboom et al. [12], cases are sorted by ascending predictive entropy; the K most-confident fraction is routed to AI and the remainder to a radiologist (modelled as oracle or 85% sensitivity). Overall sensitivity = ($\text{TP}_{\text{AI}}$ + cancers referred) / total cancers.

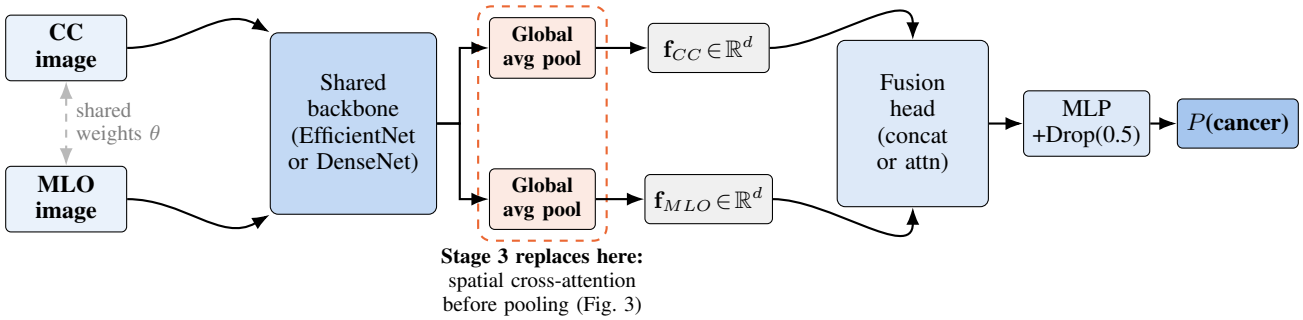



Fig. 2. BreastPairNet (Stage 2). CC and MLO images share one CNN backbone (tied $\theta$); each view is global-average-pooled (GAP) to $\mathbf{f} \in \mathbb{R}^d$, fused by concatenation or attention, and classified by a two-layer MLP with 50% dropout. The orange dashed box marks the GAP step that Stage 3 (Figure 3) replaces with spatial cross-attention *before* pooling.

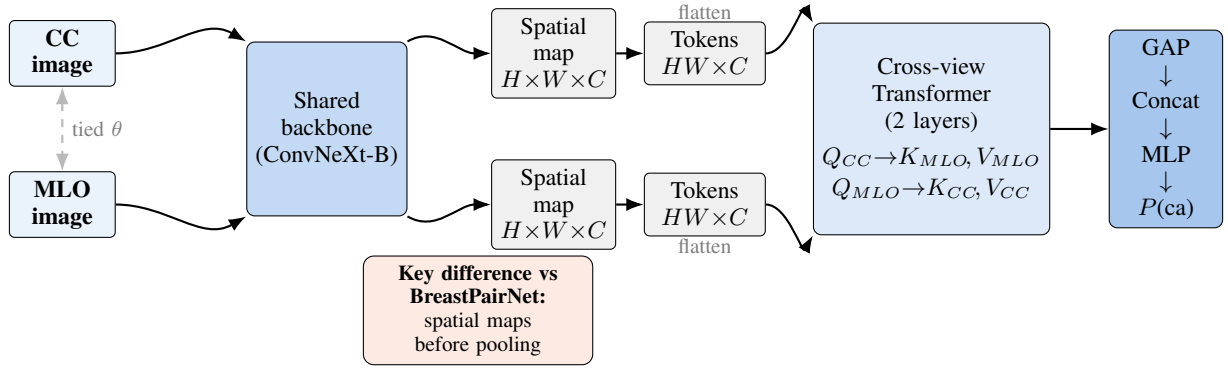



Fig. 3. CrossView Transformer (Stage 3). Spatial feature maps are flattened to tokens rather than pooled; a two-layer Transformer encoder applies bidirectional cross-attention ($Q_{CC}$ attends to $K, V$ from MLO and vice versa) so spatially consistent findings can be matched across views before pooling. Best: ConvNeXt-Base, AUROC=0.8317.

### H. Training and evaluation

All models trained on NVIDIA RTX 5070 (12 GB VRAM) with FP16 mixed-precision (AMP) and AdamW (weight decay 1e-4 for Stage 1; 1e-3 for Stage 2). Primary metric: AUROC. Also reported: F1 at Youden threshold, sensitivity, specificity, **S@90%spec** (sensitivity at 90% specificity), and **PPV** (positive predictive value, precision at the Youden-optimal threshold). 95% percentile bootstrap confidence intervals ($n$=2000 resamples) on the 2,000-pair test split (99 cancer-positive). Because all models are evaluated on the same 2,000 test pairs, paired bootstrap resampling is more statistically powerful than independent CIs for head-to-head model comparisons.

## V. Experiments and Results

### A. Stage 1 — Single-view CNN baselines

DenseNet-121 achieves the best AUROC (0.7564) but only 24.8% sensitivity at the default threshold (Table II). All five models overfit (train F1 $\approx$ 0.98) and reach only 40–45% sensitivity at 90% specificity, well below the ~85% radiologist sensitivity in screening [17].

### B. Stage 2 — Multi-view BreastPairNet

The best single model is **EfficientNet-B3 + attention + +reg** at AUROC=0.8256 (95% CI 0.7755–0.8707), a +9.1% relative gain over DenseNet-121 single-view (Table III). Each configuration was trained once; rows in Table III and Table IV report its best checkpoint re-evaluated on the test split at the Youden-optimal threshold, so the bootstrap CIs reflect test-sampling rather than run-to-run variability.

TABLE II
Single-view CNN baselines on VinDr-Mammo (n=16,000 train, n=4,000 test images).

| Architecture | Image size | Sensitivity | Specificity | F1 | AUROC | S@90% |
|---|---|---|---|---|---|---|
| VGG19 | 224 px | 0.2626 | 0.9835 | 0.3323 | 0.7066 | 0.3990 |
| ResNet-50 | 224 px | 0.2677 | 0.9847 | 0.3430 | 0.7286 | 0.4495 |
| EfficientNet-B3 | 224 px | 0.2980 | 0.9687 | 0.3138 | 0.7330 | 0.3990 |
| ResNet-50 (384 px) | 384 px | 0.2879 | 0.9821 | 0.3529 | 0.7489 | 0.4495 |
| **DenseNet-121** | **224 px** | **0.2475** | **0.9882** | **0.3356** | **0.7564** | **0.4444** |

Both best models peaked early (EfficientNet-B3 +reg at epoch 7, ConvNeXt-Base at epoch 8) and overfit thereafter. Three observations stand out (Table III). **(i) +reg consistently lifts AUROC** by from 0.008 to 0.021 across all backbone/fusion pairs. **(ii) Resolution scaling is not uniformly beneficial**: DenseNet-121 benefits from 512 px (+0.014), but EfficientNet-B3 +reg at 224 px (0.8256) beats 512 px without +reg (0.8175). **(iii) ResNet-50 degrades at 384 px** (0.7467 vs

TABLE III
BREASTPAIRNET ABLATION ACROSS BACKBONE, FUSION, AND RESOLUTION.

| Architecture | Fusion | Image | Best AUROC [95% CI] | F1 | Sens | PPV | S@90% [95% CI] |
|---|---|---|---|---|---|---|---|
| ResNet-50 | concat | 384 px | 0.7467 [0.6921–0.7998] | 0.2350 | 0.5354 | 0.151 | 0.4646 [0.3736–0.5673] |
| ResNet-50 | attention | 224 px | 0.7847 [0.7324–0.8338] | 0.2522 | 0.5859 | 0.161 | 0.5051 [0.4023–0.6019] |
| ResNet-50 | attention | 512 px | 0.7922 [0.7339–0.8429] | 0.2093 | 0.7475 | 0.122 | 0.5152 [0.4130–0.6214] |
| ResNet-50 | concat | 224 px | 0.7962 [0.7428–0.8445] | 0.2363 | 0.6970 | 0.142 | 0.4949 [0.3958–0.5977] |
| DenseNet-121 | attention | 224 px | 0.8014 [0.7487–0.8515] | 0.2800 | 0.6364 | 0.179 | 0.5758 [0.4660–0.6667] |
| EfficientNet-B3 | attention | 224 px | 0.8048 [0.7530–0.8556] | 0.2138 | 0.7677 | 0.124 | 0.5455 [0.4476–0.6449] |
| DenseNet-121 | concat | 224 px | 0.8073 [0.7577–0.8504] | 0.2101 | 0.7374 | 0.122 | 0.5152 [0.4141–0.6100] |
| DenseNet-121 **+reg** | attention | 224 px | 0.8097 [0.7594–0.8591] | 0.2033 | 0.7980 | 0.117 | 0.5556 [0.4301–0.6458] |
| DenseNet-121 | attention | 512 px | 0.8150 [0.7675–0.8617] | 0.1928 | 0.8081 | 0.109 | 0.5455 [0.4476–0.6408] |
| DenseNet-121 **+reg** | concat | 224 px | 0.8177 [0.7659–0.8649] | 0.2357 | 0.7677 | 0.139 | 0.6061 [0.5047–0.7000] |
| EfficientNet-B3 | attention | 512 px | 0.8175 [0.7698–0.8661] | 0.2807 | 0.7273 | 0.174 | 0.5556 [0.4495–0.6505] |
| **EfficientNet-B3 +reg** | **attention** | **224 px** | **0.8256 [0.7755–0.8707]** | **0.3701** | **0.6263** | **0.263** | **0.6263 [0.5229–0.7143]** |

PPV = TP/(TP+FP) at the Youden-optimal threshold. At 4.95% prevalence, PPV is inherently low; higher PPV reflects a more conservative threshold (lower sensitivity). All CIs via 2,000-resample percentile bootstrap on the 2,000-pair test split.

TABLE IV
CROSSVIEW TRANSFORMER RESULTS WITHOUT BOUNDING-BOX ANNOTATION.

| Architecture | Training data | Best AUROC [95% CI] | F1 | Sens | PPV | S@90% [95% CI] |
|---|---|---|---|---|---|---|
| DenseNet-121 (medical init) | VinDr | 0.7648 [0.7116–0.8210] | 0.2374 | 0.5960 | 0.148 | 0.5051 [0.4020–0.6100] |
| EfficientNet-B3 | VinDr | 0.7795 [0.7276–0.8300] | 0.2022 | 0.7273 | 0.117 | 0.4545 [0.3474–0.5500] |
| DenseNet-121 | VinDr | 0.7874 [0.7299–0.8401] | 0.2678 | 0.6465 | 0.169 | 0.5556 [0.4474–0.6482] |
| DenseNet-121 | VinDr+RSNA | 0.8028 [0.7484–0.8536] | 0.2640 | 0.6667 | 0.165 | 0.5354 [0.4433–0.6404] |
| **ConvNeXt-Base** | **VinDr** | **0.8317 [0.7832–0.8753]** | **0.2587** | **0.7475** | **0.156** | **0.5758 [0.4737–0.6768]** |

0.7962 at 224 px) due to batch-size reduction under VRAM pressure.

### C. Stage 3 — CrossView Transformer

*1) Performance without annotation:* **ConvNeXt-Base** achieves the best AUROC at 0.8317 (+10.0% over single-view DenseNet-121). DenseNet-121 with VinDr-Mammo+RSNA 2022 joint training reaches AUROC=0.8028, confirming incremental cross-dataset benefit. Chest-X-ray pretraining via the torchxrayvision library[1] underperforms ImageNet initialisation (AUROC 0.7648 vs 0.7874), consistent with domain mismatch between lung radiographs and mammographic texture, though tentatively, as the configurations also differ in other ways.

*2) Bounding-box annotation ablation:* **All four pairs show annotation consistently degrades AUROC by 0.122–0.201** (Table V). At epoch 4 the backbone unfreezes and fits the white box outlines as a shortcut texture; training F1 reaches ∼0.97 while test AUROC collapses because boxes are absent at test time. Possible remedies, all speculative and not yet validated for mammography, are (a) box-dropout, randomly masking boxes during training; (b) test-time box injection from a lesion detector, so test inputs match training; and (c) cropping the input to the box region instead of drawing an outline.

[1]torchxrayvision: `github.com/mlmed/torchxrayvision`

TABLE V
BOUNDING-BOX ANNOTATION ABLATION. TEST SET ALWAYS USES ORIGINAL UNANNOTATED PNGS; ΔAUROC IS RELATIVE TO THE UNANNOTATED MODEL IN TABLE IV.

| Architecture | With annotation AUROC [95% CI] | ΔAUROC |
|---|---|---|
| DenseNet-121 | 0.5862 [0.5274–0.6433] | -0.201 |
| DenseNet-121 +RSNA | 0.6675 | -0.135 |
| DenseNet-121 (medical) | 0.6433 [0.5886–0.6966] | -0.122 |
| EfficientNet-B3 | 0.6020 [0.5460–0.6617] | -0.178 |

### D. Stage 4 — MC Dropout uncertainty quantification

MC Dropout slightly improves AUROC (+0.003, within the bootstrap CI) and produces a meaningful sensitivity shift (+3.0 percentage points, Table VI). More importantly, the mean entropy is **0.5670** on cancer-positive pairs versus **0.5112** on negatives (Table VI) — a 0.056-nat gap that enables the Stage 6 triage policy.

### E. Stage 5 — Model ensemble

*1) Original ensemble (16 BreastPairNet models):* Mean-probability ensemble of 16 BreastPairNet checkpoints combined with three breast-asymmetry models and five-fold test-time augmentation (TTA×5). **Best result in the study**: AUROC=0.8458, sensitivity=0.7374, specificity=0.8296, F1=0.2944, S@90%=0.6566, S@95%=0.5657 at threshold 0.3414 (TP=73, TN=1577, FP=324, FN=26; Table VII).

TABLE VI
MC DROPOUT VS DETERMINISTIC INFERENCE (DENSENET-121/CONCAT), AND PREDICTIVE ENTROPY BY CLASS ($N = 20$ MC PASSES, 2,000 TEST PAIRS).

| Mode | AUROC | Sens | Spec | S@90% | Thr. |
|---|---|---|---|---|---|
| Deterministic | 0.8073 | 0.7374 | 0.7249 | 0.5152 | 0.3304 |
| **MC Dropout** | **0.8099** | **0.7677** | **0.7007** | **0.5253** | **0.3138** |

| Entropy | Negatives ($n = 1{,}901$) | Positives ($n = 99$) |
|---|---|---|
| Mean | 0.5112 | 0.5670 |
| Median | 0.5310 | 0.6221 |
| Std dev | 0.1436 | 0.1370 |

Gap: $+0.056$ nats. Higher entropy $\Rightarrow$ refer high-uncertainty cases.

TABLE VII
BEST ENSEMBLE RESULT (MEAN PROBABILITY, TTA$\times$5) ON VINDR-MAMMO (2,000 PAIRS, 99 CANCERS) AND ZERO-SHOT ON RSNA 2022 (4,764 PAIRS, 110 CANCERS).

| Metric | VinDr-Mammo | RSNA 2022 (zero-shot) |
|---|---|---|
| AUROC | 0.8458 | 0.6008 [0.5577–0.6585] |
| Sensitivity | 0.7374 | 0.3909 |
| Specificity | 0.8296 | 0.7598 |
| F1 | 0.2944 | 0.0677 |
| PPV | 0.184 | 0.037 |
| S@90% | 0.6566 | 0.1636 |
| S@95% | 0.5657 | 0.1182 |
| Threshold | 0.3414 | 0.4823 |

PPV=0.184 means 1 in 5.4 recalls is cancer-positive at this threshold; at 4.95% test prevalence, this is consistent with reported breast-cancer AI triage literature (PPV 15–25%).
RSNA 2022 threshold-dependent metrics use the Youden-optimal threshold on RSNA 2022 itself (0.4823).

This ensemble does **not** include ConvNeXt-Base or EfficientNet-B3 +reg models.

*2) Ensemble variants:* Two variants were also evaluated: adding all CrossView checkpoints (25 models, including lower-epoch snapshots) gives AUROC=0.8441, S@90%=0.6566 and sensitivity=0.7677, and a selective ensemble (top-3 +reg, top-5 CrossView and 3 asymmetry models; 11 models) gives AUROC=0.8446 and S@90%=0.6667. Across the three configurations, AUROC differences (0.8441–0.8458) are within the $\pm 0.04$ percentile bootstrap CI; the selective ensemble offers a marginal S@90% benefit (0.6667 vs 0.6566). Full metrics (specificity, F1, threshold) were only computed for the primary 16-model configuration; secondary configurations were evaluated on AUROC and the clinical operating point (S@90%) only.

### F. Stage 6 — Hybrid triage simulation

Under oracle assumptions, **K=50% delivers 87.9% overall sensitivity with 50% workload reduction** (Table VIII). Under realistic 85% radiologist sensitivity [17], the same K=50% yields 77.3% — below the clinical floor. Achieving $\geq$80% overall sensitivity under realistic conditions requires K$\leq$40% (Table VIII).

TABLE VIII
HYBRID TRIAGE SWEEP (ORACLE VS. 85% RADIOLOGIST SENSITIVITY).

| Frac | Wkld $\Delta$ | Sen (or.) | Sen (85%r) | AI Sen | Recall |
|---|---|---|---|---|---|
| 30% | -30% | 0.9495 | 0.8253 | 0.7059 | 0.0017 |
| 40% | -40% | 0.9192 | 0.8071 | 0.6800 | 0.0013 |
| **50%** | **-50%** | **0.8788** | **0.7727** | **0.5862** | **0.0031** |
| 60% | -60% | 0.8283 | 0.7359 | 0.5526 | 0.0052 |
| 70% | -70% | 0.7677 | 0.6935 | 0.5400 | 0.0133 |
| 80% | -80% | 0.7677 | 0.7207 | 0.6618 | 0.1305 |
| 90% | -90% | 0.7677 | 0.7450 | 0.7262 | 0.2238 |

### G. Comparison to literature

Our best ensemble (AUROC=0.8458) exceeds MamT4 [18] (AUROC=0.840), but MamT4 excludes BI-RADS 3 cases from evaluation, while our pipeline retains them as negative, making the AUROC values not directly comparable. McKinney et al. [11] (AUC=0.889) and Yoon et al. [4] (pooled AUC=0.87) report on different datasets and are likewise not directly comparable.

### H. External validation on RSNA 2022

The best ensemble (AUROC=0.8458 on VinDr-Mammo) was applied zero-shot to RSNA 2022. Full metrics are shown in Table VII.

Three factors may drive the gap: (1) Vietnamese vs United States breast density distribution and lesion prevalence [1], [11]; (2) different FFDM equipment and post-processing pipelines; (3) the S@90%spec drop ($-75.1\%$) is particularly stark at the clinical operating point, suggesting the conservative side of the decision surface is most affected by domain shift.

## VI. DISCUSSION

**Multi-view fusion.** The +9.1% AUROC gain from BreastPairNet is consistent with lesions appearing unambiguously in only one projection; joint CC/MLO conditioning likely reduces view-specific false positives, though the contributions of shared weights and of the fusion architecture cannot be separated here.

**Regularisation benefit.** With only $\sim$395 cancer-positive training pairs, stronger augmentation (elastic deformation, mixup, label smoothing) likely acts as a powerful implicit regulariser.

**CrossView Transformer vs BreastPairNet.** The two are not statistically separated (see below). One plausible explanation for the small CrossView advantage is that exchanging spatial feature maps before global pooling may preserve localisation information that BreastPairNet's independent pooling discards, though confounded by the different backbone architectures.

**Bounding-box shortcut learning.** Painted boxes are a texture shortcut perfectly correlated with the label in training but absent at test time, consistent with well-documented shortcut learning in medical imaging.

**Why adding the highest-AUROC model to the ensemble can reduce AUROC.** Adding ConvNeXt-Base (the highest

individual AUROC, 0.8317) to the 16-model BreastPairNet ensemble lowers AUROC from 0.8458 to 0.8441, because ensemble gains come from *error diversity*, not from member rank. ConvNeXt-Base shares the CrossView architecture with the DenseNet-121 and EfficientNet-B3 CrossView members already present, so its errors are correlated with theirs and add no new coverage of failure modes. Mean-probability pooling is also sensitive to miscalibration: a systematically over- or under-confident model can shift the pooled mean away from the ensemble's Youden-optimal threshold, reducing AUC even when its own ranking is strong. The selective ensemble partly mitigates this by favouring diversity (three +reg BreastPairNet, five CrossView, three asymmetry models) over the single best-ranked model.

**PPV and triage utility.** PPV at the Youden threshold ranges from 0.109 to 0.263 (Stage 2) and 0.117 to 0.169 (Stage 3), consistent with the 4.95% positive prevalence; the ensemble's 0.184 is discussed with Table VII. Screening prevalence is usually lower than VinDr-Mammo's, so deployed PPV would be lower, and calibrated outputs with prevalence-adjusted thresholds would be needed before clinical use.

**Statistical comparisons and paired testing.** With only 99 cancer-positive test pairs, the 95% percentile bootstrap CI width is $\pm$0.04–0.05 AUROC, so *adjacent-stage* differences, namely Stage 2 vs Stage 3 ($\Delta = 0.006$), Stage 3 vs ensemble ($\Delta = 0.014$), and BreastPairNet vs CrossView Transformer (95% CIs 0.776–0.871 vs 0.783–0.875), cannot be claimed as individually significant. The percentage gains (+9.1%, +10.0%, +11.8%) are observed AUROC changes to be read alongside the CIs, not confirmed step-wise improvements.

Under *paired* bootstrap resampling on the same 2,000 test pairs, the ensemble's cumulative gain over Stage 1 ($\Delta = 0.089$ AUROC) is expected to reach statistical separation, whereas adjacent-stage gains, an order of magnitude smaller, are unlikely to; formal paired tests will be reported with the retrained models. The key claim is therefore the cumulative gain over single-view baselines, and the ablation of which components help or hurt, not that each stage beats the last.

**Contribution relative to MamT4.** Beyond the evaluation difference noted in the comparison to literature with MamT4 [18] (AUROC=0.840), MamNet-6 contributes a controlled ablation study that isolates the effect of multi-view fusion, regularisation, spatial cross-attention, and ensemble size—components that MamT4 does not systematically ablate. MamNet-6 also adds uncertainty quantification (Stage 4) and a hybrid triage simulation (Stage 6) grounded in a published clinical framework [12], enabling a workload-reduction analysis that MamT4 does not provide.

**Limitations.** (i) Small positive test count (99 pairs) limits fine-grained comparison; (ii) large domain gap to RSNA 2022; (iii) 7,054 non-CC/MLO RSNA images excluded from pair construction; (iv) operating-point metrics (sensitivity, specificity, F1, PPV) use Youden-optimal thresholds chosen on the evaluation data and are therefore optimistic, whereas AUROC, S@90% and S@95% are threshold-independent.

**Future work.** Box-dropout or test-time box injection; CrossView at 512 px; validation on CBIS-DDSM and INbreast; a tighter +reg/CrossView/asymmetry ensemble; RSNA fine-tuning; and a prospective triage study with real reader sensitivity.

## VII. Conclusion

We have presented MamNet-6, a six-stage pipeline for breast cancer detection on VinDr-Mammo that rises from a single-view AUROC of 0.7564 to 0.8458 for the 16-model ensemble (+11.8%). Bounding-box annotations consistently degraded AUROC by 0.12–0.20 due to shortcut learning—the most actionable negative finding. MC Dropout's 0.056-nat entropy gap enables hybrid triage: 50% AI reads preserve 87.9% sensitivity with an oracle reader but 77.3% with an 85%-sensitive reader, so K$\leq$40% is needed to stay above the 80% floor. External validation on RSNA 2022 (AUROC=0.6008) shows that generalisation to United States FFDM remains a critical open challenge.

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